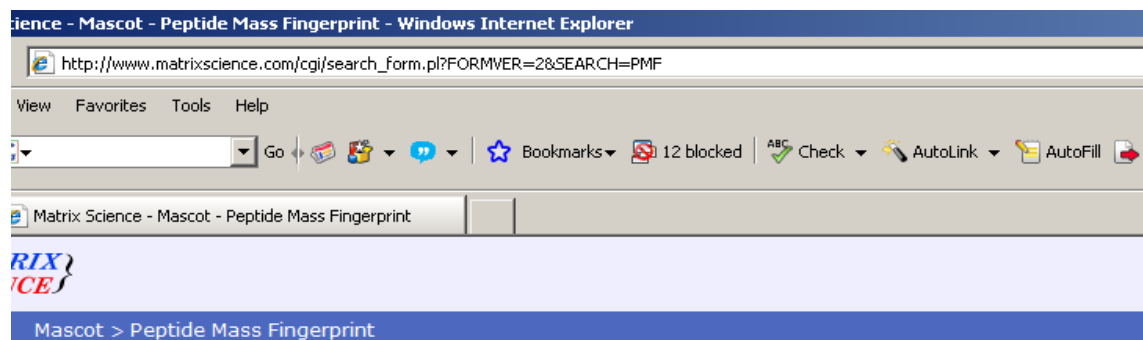


**PMF hands-on practice** (please use PC not Mac):

3.1. **Instructional and Practice only:** Please go to FingerPrint Library (FPLibrary) and practice with Mass Spectra Examples from Rice Pollen Grain  
<http://www.moleculardetective.org/FPLibrary/FPLibraryMenu.html>

**Here are the steps:**

- Download and run the Moverz Bioinformatics Software
- Practice with Mass Spectra Examples from Rice Pollen Grain.  
Please use Internet Explorer  
if it does not open the mass fingerprints, please put mouse cursor underneath the Example file, right-click to "Save Target As" onto your desktop. The file format should be "Moverz document"
- Run Moverz software, open the \*.massml file you downloaded.
- Practice delete all masses and autolabel peaks (A). Inspect isotope envelopes and peptide charge states. If you mess up the spectrum, you can always go back to the library and download again.
- Go to Edit, and copy all the masses.
- Go back to the main page and click on "Protein Search Engine"
- Put in your name, e-mail and search title (your sample name).
- Choose NCBI nr database or the other databases for testing
- Since these fingerprints are from plants, so select "Viridiplantae (Green plants).
- Make sure the following parameters:



## MASCOT Peptide Mass Fingerprint

|                                                                                    |                                                                                                                                                                                                                                                 |                                           |                                                 |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|
| <b>Your name</b>                                                                   | <input type="text" value="SXC"/>                                                                                                                                                                                                                | <b>Email</b>                              | <input type="text" value="schen@ufl.edu"/>      |
| <b>Search title</b>                                                                | <input type="text"/>                                                                                                                                                                                                                            |                                           |                                                 |
| <b>Database</b>                                                                    | <input type="text" value="NCBIInr"/>                                                                                                                                                                                                            |                                           |                                                 |
| <b>Taxonomy</b>                                                                    | <input type="text" value="... Viridiplantae (Green Plants)"/>                                                                                                                                                                                   |                                           |                                                 |
| <b>Enzyme</b>                                                                      | <input type="text" value="Trypsin"/>                                                                                                                                                                                                            | <b>Allow up to</b>                        | <input type="text" value="1"/> missed cleavages |
| <b>Fixed modifications</b>                                                         | <input type="text" value="Acetyl (K)"/> <input type="text" value="Acetyl (N-term)"/> <input type="text" value="Acetyl (Protein N-term)"/> <input type="text" value="Amidated (C-term)"/> <input type="text" value="Amidated (Protein C-term)"/> |                                           |                                                 |
| <b>Variable modifications</b>                                                      | <input type="text" value="Acetyl (K)"/> <input type="text" value="Acetyl (N-term)"/> <input type="text" value="Acetyl (Protein N-term)"/> <input type="text" value="Amidated (C-term)"/> <input type="text" value="Amidated (Protein C-term)"/> |                                           |                                                 |
| <b>Protein mass</b>                                                                | <input type="text"/> kDa                                                                                                                                                                                                                        | <b>Peptide tol. ±</b>                     | <input type="text" value="0.3"/> Da             |
| <b>Mass values</b>                                                                 | <input checked="" type="radio"/> MH <sup>+</sup> <input type="radio"/> M <sub>r</sub> <input type="radio"/> M-H <sup>-</sup>                                                                                                                    |                                           |                                                 |
| <b>Monoisotopic</b>                                                                | <input checked="" type="radio"/> Average <input type="radio"/>                                                                                                                                                                                  |                                           |                                                 |
| <b>Data file</b>                                                                   | <input type="text"/> <input type="button" value="Browse..."/>                                                                                                                                                                                   |                                           |                                                 |
| <b>Query</b><br>NB Contents of this field are ignored if a data file is specified. | <input type="text"/>                                                                                                                                                                                                                            |                                           |                                                 |
| <b>Decoy</b>                                                                       | <input type="checkbox"/>                                                                                                                                                                                                                        |                                           |                                                 |
| <b>Report top</b>                                                                  | <input type="text" value="AUTO"/> hits                                                                                                                                                                                                          |                                           |                                                 |
| <input type="button" value="Start Search ..."/>                                    |                                                                                                                                                                                                                                                 | <input type="button" value="Reset Form"/> |                                                 |

- Paste the masses to the query window.
- Click on Start Search.
- Evaluate whether the identification is confident by examining the Score and the peptide matching result, e.g., number of peptides matched, score, sequence coverage, and mass accuracy etc.
- Make sure you understand the principle of the Peptide Mass Fingerprinting.

### 3.2. Questions (4 points):

You are a biosleuth. Please go to FPLibrary on the above website. Please identify with high confidence **ONE of the rice (*Oryza sativa*) proteins** using “FBI fingerprint database” and the “Mass Spectra Fingerprints from Customers”.

- 1)** Please write a nice report of each ID with the protein name (see below for an example, just screen capture from the Mascot output showing the confident parameters).
- 2)** For your ID above, please briefly describe how you judge the confidence of the IDs.

Here is an example of the report (this is just an example).

## Protein ID

Fingerprint 1, Confident ID: Os10g0462900, peroxidase.

